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Question 1

Not answered

Marked out of 3.00

Multivariate techniques

Choose one option from the given list for each description below:

A researcher wants to analyze the various chains of influence existing among degree of poverty, social isolation, unemployment, depression and physical morbidity.

A researcher wants to predict the value that a binary dependent variable can take using an independent variable and a constant.

Within subjects analysis for paired observations of means in 5 different groups.

Your answer is incorrect.

Explanation:

When the term 'chain of causation' is used, path analysis is often a suitable technique.

To make predictions, we need regression models. As a binary dependent variable is employed, logistic regression is the most appropriate.

Repeated measures ANOVA can be used to compare paired observations within subjects across multiple groups. Here one will model the groups (5 groups) as a between subjects factor and time of observation (before and after) as the within-subjects factor.

The correct answer is: A researcher wants to analyze the various chains of influence existing among degree of poverty, social isolation, unemployment, depression and physical morbidity. → Path analysis, A researcher wants to predict the value that a binary dependent variable can take using an independent variable and a constant. → Logistic regression, Within subjects analysis for paired observations of means in 5 different groups. → Repeated measures ANOVA

Question 2

Not answered

Marked out of 3.00

Univariate statistics

Choose one option from the given list for each description below:

A researcher wants to assess the strength of association between educational attainment rated as a continuous measure and degree of compliance rated as a continuous measure.

A researcher wants to compare the proportion of smoking psychiatrists and nonsmoking psychiatrist in a local hospital who has been off sick at least one day in last 28 days.

The degree of agreement and reliability between two raters for nominal measures.

Your answer is incorrect.

Explanation: To measure the strength of association between two continuous measures, correlation can be tested using Pearson's r. Chi-square test is useful to compare proportions between two groups. Reliability of nominal measures can be tested using kappa.

The correct answer is: A researcher wants to assess the strength of association between educational attainment rated as a continuous measure and degree of compliance rated as a continuous measure. → Pearson's test, A researcher wants to compare the proportion of smoking psychiatrists and nonsmoking psychiatrist in a local hospital who has been off sick at least one day in last 28 days. → Chi-square test, The degree of agreement and reliability between two raters for nominal measures. → Kappa

Question 3

Not answered

Marked out of 5.00

Sampling methods I

Choose the most appropriate sampling technique used for each study described below:

An addiction psychiatrist is keen to investigate the effect of DVLA warning on heroin users. He identifies four regular patients in his methadone clinic as suitable for the study and distributes copies of screening questionnaire through them to other heroin users known to drive a car.

To study the amount of over-the-counter use of analgesics in depressed patients, first 50 consecutive patients attending out patient clinic are screened.

To study the amount of over-the-counter use of analgesics in depressed patients, a questionnaire is sent to every 10th patient in an out-patient register between two specified periods of time.

In order to study the amount of over-the-counter use of analgesics in depressed patients, patients in an outpatient register are grouped by sex, age >65 or <65, presence or absence of a history of analgesic prescription. A random sample is picked from each group.

In order to study the amount of over-the-counter use of analgesics in depressed patients, patients in an outpatient register are grouped by sex, age >65 or <65, presence or absence of a history of analgesic prescription. First ten patients from each group are picked.

Your answer is incorrect.

Explanation:

Snowball sampling is used to reach normally difficult-to-engage sample frame.

Consecutive sampling is a form of convenience sampling.

Using every nth person as if in roll calls produces a systematic sample

Using every 10th person, as in the red cars, produces a systematic sample.

After stratifying a target population into different strata, random samples are picked from each stratum in question 4 - this constitutes a stratified random sampling.

In question 5, after the stratification, a proportion is selected - this is called quota sampling.

The correct answer is: An addiction psychiatrist is keen to investigate the effect of DVLA warning on heroin users. He identifies four regular patients in his methadone clinic as suitable for the study and distributes copies of screening questionnaire through them to other heroin users known to drive a car. → Snowball sampling, To study the amount of over-the-counter use of analgesics in depressed patients, first 50 consecutive patients attending out patient clinic are screened. → Convenience sampling, To study the amount of over-the-counter use of analgesics in depressed patients, a questionnaire is sent to every 10th patient in an out-patient register between two specified periods of time. → Systematic sampling, In order to study the amount of over-the-counter use of analgesics in depressed patients, patients in an outpatient register are grouped by sex, age >65 or <65, presence or absence of a history of analgesic prescription. A random sample is picked from each group. → Stratified random sampling, In order to study the amount of over-the-counter use of analgesics in depressed patients, patients in an outpatient register are grouped by sex, age >65 or <65, presence or absence of a history of analgesic prescription. First ten patients from each group are picked. → Quota sampling



Question 4

Not answered

Marked out of 4.00

Statistical tests II

Hyperglycemia and metabolic syndrome are associated with the use of antipsychotics in schizophrenia patients. A researcher wants to investigate this phenomenon for the newly marketed antipsychotic paliperidone. He has various options with respect to study designs. For each design below choose ONE appropriate statistical test from the given list.

The mean increase in glucose levels after using antipsychotic paliperidone for eight weeks in a sample of 50 inpatients.

The mean change in glucose levels after using antipsychotics for eight weeks in a sample of 50 inpatients taking antipsychotic paliperidone compared to 33 patients attending day unit for psychotherapy for the same period.

An increase in the prevalence of metabolic syndrome in a sample of 50 inpatients after using antipsychotic paliperidone for eight weeks.

Comparing the prevalence of metabolic syndrome in a sample of 50 inpatients after using antipsychotic paliperidone for eight weeks and a sample of 33-day unit patients undertaking psychotherapy for eight weeks.

Your answer is incorrect.

Explanation:

For question 1, mean values before and after are compared in the same sample: paired t-test can be used for this.

For question 2, differences in values of each group are first computed, then a mean is obtained for each group which is then compared: this is effectively the same as comparing two means which are obtained by other means. Hence, we can use unpaired t test.

Question 3 refers to comparing proportions in a paired condition (before and after) - McNemar's test is best used for this purpose. It is similar to Chi-square test but can be used in non-parametric or paired situations.

Question 4 is comparing proportions between two independent groups - Chi-square is the most suitable test.

The correct answer is: The mean increase in glucose levels after using antipsychotic paliperidone for eight weeks in a sample of 50 inpatients. → Paired t-test, The mean change in glucose levels after using antipsychotics for eight weeks in a sample of 50 inpatients taking antipsychotic paliperidone compared to 33 patients attending day unit for psychotherapy for the same period. → Students t-test, An increase in the prevalence of metabolic syndrome in a sample of 50 inpatients after using antipsychotic paliperidone for eight weeks. → McNemar's test, Comparing the prevalence of metabolic syndrome in a sample of 50 inpatients after using antipsychotic paliperidone for eight weeks and a sample of 33-day unit patients undertaking psychotherapy for eight weeks. → Chi-square test



Question 5

Not answered

Marked out of 4.00

Statistical tests III

For each of the studies below, choose the most appropriate statistical test to analyse the data.

Comparison of systolic blood pressures in independent samples of schizophrenia patients taking clozapine with patients taking olanzapine.

Comparison of the prevalence of hepatitis C among IV heroin users and IV cocaine users.

Comparison of the level of platelet serotonin in depressed adolescents and adults following administration of three different antidepressant drugs.

Comparison of serum cholesterol before and after attending a lifestyle modification clinic in a group of bipolar patients attending assertive outreach service.

Your answer is incorrect.

Explanation:

Comparison of mean blood pressure between two groups can be achieved by employing Student's unpaired t test. Prevalence is a proportion.

Comparison of proportions (categorical outcome) can be achieved using chi-square test.

Comparing means of a continuous variable such as platelet serotonin level in more than two groups can be done using ANOVA. Please note that while ANOVA indicates whether or not differences exist among groups, it does not tell where such differences are located.

Paired observations of mean values (serum cholesterol in the same subject before and after an intervention) can be tested using paired t-test.

The correct answer is: Comparison of systolic blood pressures in independent samples of schizophrenia patients taking clozapine with patients taking olanzapine. → Student t-test, Comparison of the prevalence of hepatitis C among IV heroin users and IV cocaine users. → Chi-square test., Comparison of the level of platelet serotonin in depressed adolescents and adults following administration of three different antidepressant drugs. → Analysis of variance, Comparison of serum cholesterol before and after attending a lifestyle modification clinic in a group of bipolar patients attending assertive outreach service. → Paired t-test



Question 6

Not answered

Marked out of 4.00

Sampling methods II

For each of the techniques described below, choose the most appropriate sampling technique from the given list.

Choosing volunteers from a trust induction seminar to participate in a survey of the prescribing practice of psychiatrists.

Listing individuals in a locality by ethnic group and choosing a proportion from each ethnic group at random.

Initially, numbering all the constituents of a sampling frame and then using a random number table to pick cases.

Randomly selecting deaneries, and then sampling every trainee attached to a deanery.

Your answer is incorrect.

Explanation:

A convenience sample is a sample where the patients are selected, in part or whole, at the convenience of the researcher. The researcher makes no or limited attempt to ensure representativeness of this sample to the source population. A good example of a convenience sample is standing at a shopping mall and selecting shoppers as they walk by to fill out a survey on shopping practices.

In simple random sampling each individual in the population has an equal chance (probability) of being selected.

Quota sampling refers to stratification of source population into subgroups and allocating a proportion to each group; then in a non-random manner, (e.g. first 10 in each group) the individuals are selected to be included in the study. Both convenience sampling and quota sampling are non-random methods. One way of obtaining a random sample is to give each individual a number and then to use a computer-generated table of random numbers to decide which individuals should be included.

Cluster sampling involves the use of groups as the sampling unit rather than individuals (e.g. households, school classes or GP practices). The groups to be studied should be randomly selected from all possible groups of the same type. Most importantly, all members of the selected groups must be included in the study to qualify as a cluster sample. If individuals are again randomly selected within the random clusters, this will be a multi-stage sample (two random sampling). In stratified sampling, a population is divided into subgroups based on defined characteristics such as age, sex, or severity of illness, or any combination of these; then random samples are selected from each subgroup. In such sample, the probability of an individual being included varies according to a predetermined characteristic (strata). This method ensures that small subgroups that are of particular interest to the investigator are adequately represented.

The correct answer is: Choosing volunteers from a trust induction seminar to participate in a survey of the prescribing practice of psychiatrists. → Convenience sampling, Listing individuals in a locality by ethnic group and choosing a proportion from each ethnic group at random. → Stratified random sampling, Initially, numbering all the constituents of a sampling frame and then using a random number table to pick cases. → Simple random sampling, Randomly selecting deaneries, and then sampling every trainee attached to a deanery. → Cluster sampling



Question 7

Not answered

Marked out of 3.00

Sampling methods III

For each of the techniques described below, choose the most appropriate sampling technique from the given list.

Selecting all persons in a randomly selected GP practice.

Selecting every 4th outpatient in the register

A sample is categorised according to identified mediators and effect modifiers, and subjects are randomly picked within each category to constitute study sample.

Your answer is incorrect.

Explanation:

The difference between cluster sampling and simple random sampling is in the individual unit that is picked randomly. If large units such as GP practices, postcode areas or classrooms are picked wherein all individual members of the group are studied then it is called **cluster sampling**. In simple random samples, the individual participants are directly picked in a random fashion. In **stratified random samples**, larger subgroups are identified according to known mediators, and in each group, members are picked randomly. Hence the groups are not randomly chosen, unlike cluster sampling and within each group not all members are included, again in contrast from cluster sampling.

Systematic sampling appears organised and 'systematic' where every nth member is picked for a study. But the probability of being chosen is not random and is influenced by unknown factors. Hence, it is a non-random method of sampling.

The correct answer is: Selecting all persons in a randomly selected GP practice. → Cluster sampling, Selecting every 4th outpatient in the register → Systematic sampling, A sample is categorised according to identified mediators and effect modifiers, and subjects are randomly picked within each category to constitute study sample. → Stratified sampling

Question 8

Not answered

Marked out of 3.00

Errors in research

For each of the errors described below, choose the most appropriate term from the given list.

A religious leader quotes epidemiological data that countries with the highest use of condoms and other artificial methods of contraceptive practices have a higher risk of AIDS. He asserts that condoms are ineffective in reducing HIV transmission.

A researcher fails to show the effectiveness of clozapine in treatment-resistant schizophrenia despite conducting a rigorous randomised controlled trial for 18 months. This null-result is in contrast to popular belief and prior research evidence.

In a case-control study of autism and parental bonding, gender matched siblings of same households are chosen as controls. Despite having a high prevalence of poor bonding in autistic children, the differences do not reach statistical significance.

Your answer is incorrect.

Explanations:

In ecological studies risk factors are not examined in each individual; instead prevalence of risk factors in a large (often naturally occurring) group such as councils, counties, districts, countries, etc are associated with prevalence of an outcome in the same unit. This may produce erroneous results as the individual with a risk factor may not be the same individual who suffers the outcome. This is called ecological fallacy. This question is a classical example of the **ecological fallacy**.

A common reason for non-replication of widely established findings is **type 2 error** - this may be due to lack of sufficient sample size for the experiment.

Selection bias occurs when the subjects selected for the study are not representative of the population they come from. Selecting controls in analytical studies must be carefully done to avoid this bias. In this study the trouble with selecting siblings for controls is that sibling of an

autistic child is much more likely to have poor bonding either due to genetic influences or parenting styles than a child from the general population. Thus poor bonding will be overrepresented among the controls too, and results might make the association appear weaker than the reality.

The correct answer is: A religious leader quotes epidemiological data that countries with the highest use of condoms and other artificial methods of contraceptive practices have a higher risk of AIDS. He asserts that condoms are ineffective in reducing HIV transmission. → Ecological fallacy, A researcher fails to show the effectiveness of clozapine in treatment-resistant schizophrenia despite conducting a rigorous randomised controlled trial for 18 months. This null-result is in contrast to popular belief and prior research evidence. → Type 2 error, In a case-control study of autism and parental bonding, gender matched siblings of same households are chosen as controls. Despite having a high prevalence of poor bonding in autistic children, the differences do not reach statistical significance. → Selection bias



Question 9

Not answered

Marked out of 3.00

Data Types

For each of the following descriptions choose an appropriate response from given list:

A prospective study investigates relapse rates of depression in patients who have maintenance ECT.

A neuroimaging study measures insight using a single item as good, moderate, mildly abnormal, poor or absent to correlate with parietal volume.

Lean body mass on admission at an eating disorder unit.

Your answer is incorrect.

Explanation:

Having a relapse or not is a categorical outcome (i.e. 'yes' or 'no' type). Hence, the outcome is **nominal**. Likert scales, as described here in the insight measurement, produce **ordinal** data. Lean body mass is a continuous variable with a meaningful zero. So one can say that subject A is two times heavier than subject B. As such ratios are possible, lean body mass is a **ratio** data. Consider **interval** scales such as dates of admission to a psychiatric unit. These are ordered and have real numerical units and can be subtracted from each other, but there is no intrinsic zero to the scale so that dividing them does not make sense. For example, it would be absurd to say one date of admission is three times 'bigger' than another date.

The correct answer is: A prospective study investigates relapse rates of depression in patients who have maintenance ECT. → Binary Nominal, A neuroimaging study measures insight using a single item as good, moderate, mildly abnormal, poor or absent to correlate with parietal volume. → Ordinal, Lean body mass on admission at an eating disorder unit. → Ratio

Question 10

Not answered

Marked out of 5.00

Graphs describing data

Choose one appropriate option from the given list for each of the following descriptions:

A graph where both vertical and horizontal columns are made of numbers where all individual data are represented instead of displaying bars of various colours to represent the magnitude.

A graph where a vertical line extends on either side of a vertical bar displaying interquartile range.

A graph where two continuous variables are plotted after paired observations resulting in numerous dots.

A graph used in reporting meta-analyses where in the effect size is represented by a diamond-shaped lozenge.

A graph that could be used to identify potential heterogeneity in which effect size is plotted against a measure of precision.

Your answer is incorrect.

Explanations: The answers are self-explanatory. L'Abbe plot is also used in a meta-analysis. The proportion of subjects improved with placebo is plotted on one x-axis; proportion improved on active medication is plotted on the y axis. The individual trials are often indicated by differently sized symbols.

<http://www.medicine.ox.ac.uk/bandolier/booth/glossary/labbe.html>

The correct answer is: A graph where both vertical and horizontal columns are made of numbers where all individual data are represented instead of displaying bars of various colours to represent the magnitude. → Stem and leaf plot, A graph where a vertical line extends on either side of a vertical bar displaying interquartile range. → Box-Whisker plot, A graph where two continuous variables are plotted after paired observations resulting in numerous dots. → Scattergram, A graph used in reporting meta-analyses where in the effect size is

represented by a diamond-shaped lozenge. → Forest plot, A graph that could be used to identify potential heterogeneity in which effect size is plotted against a measure of precision. → Galbraith plot



Question 11

Not answered

Marked out of 3.00

Measures of Agreement

	Nurse		
Psychiatrist	Borderline +	Borderline -	Total
Borderline +	40	20	60
Borderline -	10	30	40
Total	50	50	100

Using the given table answer the following questions on measures of agreement

The overall agreement between the two clinicians out of the total

The overall agreement by chance alone

The value of kappa

Your answer is incorrect.

Explanation:

Step 1:

Number of occasions of agreement = 40 (agreed for presence of diagnosis)+30 (agreed for absence of diagnosis) = 70 out of 100 individual events = 70% . This is the overall percent agreement between the two clinicians out of the total.

Step 2:

To determine the effect of chance, assume that the psychiatrist has no clinical skills and, as a result, has a tendency to depend on his stare

To determine the effect of chance, assume that the psychiatrist has no clinical skills and, as a result, has a tendency to depend on his stars all the time. If this is the case, he must be diagnosing 60 out of any 100 as borderline and 40 out of any 100 as not having borderline irrespective of the population (from the above data). If he evaluates the 50 patients diagnosed as borderline by the nurse, he must call 30 of them to have the diagnosis; using the same principles he must call 20 out of 50 patients rejected by the nurse as non-borderline. So if he depends on chance purely, he must agree with the nurse on $30+20 = 50$ out of 100 times. The overall percent agreement by chance alone must be 50%.

Step 3:

Kappa expresses the extent to which the observed agreement exceeds that which would be expected by chance alone (numerator) relative to the most that the observers could hope to improve their agreement (i.e., $100\% - \text{agreement expected by chance alone}$) [denominator].
 $= (70-50) / (100-50) = 20/50 = 0.4$

The correct answer is: The overall agreement between the two clinicians out of the total $\rightarrow 0.7$, The overall agreement by chance alone $\rightarrow 0.5$, The value of kappa $\rightarrow 0.4$



Question 12

Not answered

Marked out of 3.00

Numbers in critical appraisal I

For each of the questions below, select the one most appropriate number from the list given.

The point at which the confidence interval for a difference between means becomes non-significant.

The likelihood ratio for a negative test when sensitivity is 60%, and specificity is 80%.

The correlation coefficient where there is an absolute inverse association between two variables.

Your answer is incorrect.

Explanation:

Q1. When the confidence interval range includes the value 0 for mean differences, the results become insignificant. The same applies to a range of ratios when they include the value 1.

Q2. The likelihood ratio of negative test = $[(1 - \text{sensitivity}) / \text{specificity}] = [(1 - 60\%) / 80\%] = [(1 - 0.6) / 0.8] = 0.4 / 0.8 = 1/2 = 0.5$

Q3. A correlation coefficient of -1 means there is an absolute inverse association between the two variables

The correct answer is: The point at which the confidence interval for a difference between means becomes non-significant. → 0, The likelihood ratio for a negative test when sensitivity is 60%, and specificity is 80%. → 0.5, The correlation coefficient where there is an absolute inverse association between two variables. → -1

Question 13

Not answered

Marked out of 4.00

Sampling methods IV

Choose one option for each question below.

A researcher is interested in measuring dimensional factors of PTSD in illegal immigrants. He could not find a reliable sampling frame for recruitment.

Choose...

A researcher is interested in studying the lifestyle factors influencing metabolic syndrome in patients who receive antipsychotic prescriptions. He classifies the eligible sample into inpatients, outpatients and CMHT clients before recruiting for his study, to balance for the effect of one's environment on metabolic syndrome.

Choose...

A research team is interested in studying the prevalence of depression across all primary care attendees in The UK. But due to the scope of the task, they decide to pick randomly representative primary care authorities and survey all patients in the selected regions.

Choose...

A research nurse wants to study the experience of some mentally ill parents with respect to the demands of their teenage children. She designs a qualitative study method to carry out this work.

Choose...

Your answer is incorrect.

Explanation:

Question 1: To locate difficult to recruit individuals such as in the question, one can use snowballing technique; here one participant identifies another participant, who recruits one more and so on.

In question 2, as the existing sample is classified into various groups (strata) to enable equitable distribution of a possible mediating factor, this is stratified random sampling.

Instead of carrying out a whole country census, the researchers have randomly picked GP clusters and included everyone in the clusters in question 3. Instead, if they randomly picked individuals after randomly picking clusters, this would have been multistage sampling.

Purposive samples are suitable for qualitative study in which required outliers could be conveniently picked, as in question 4.

The correct answer is: A researcher is interested in measuring dimensional factors of PTSD in illegal immigrants. He could not find a reliable sampling frame for recruitment. → Snowball sampling, A researcher is interested in studying the lifestyle factors influencing metabolic syndrome in patients who receive antipsychotic prescriptions. He classifies the eligible sample into inpatients, outpatients and CMHT clients before recruiting for his study, to balance for the effect of one's environment on metabolic syndrome. → Stratified random sampling, A research team is interested in studying the prevalence of depression across all primary care attendees in The UK. But due to the scope of the task, they decide to pick randomly representative primary care authorities and survey all patients in the selected regions. → Cluster sampling, A research nurse wants to study the experience of some mentally ill parents with respect to the demands of their teenage children. She designs a qualitative study method to carry out this work. → Convenience sampling



Question 14

Not answered

Marked out of 3.00

Glossary of statistical terms 1

For each description below choose the most appropriate answer from the given list.

The ability to detect the true difference, if it exists, using the given sample.

Detecting a difference between studied groups when such difference does not exist.

The probability that a range of specified values contains the true value of the variable.

Your answer is incorrect.

Explanations:

Q1. The power of a study is its ability to detect a true difference if it exists.

Question 2 describes a type I error. This may be due to the high threshold for p-value (greater than usual 0.05) or multiple testing.

Question 3 refers to confidence intervals. The confidence level is usually set at 95%.

The correct answer is: The ability to detect the true difference, if it exists, using the given sample. → Power, Detecting a difference between studied groups when such difference does not exist. → Type 1 error, The probability that a range of specified values contains the true value of the variable. → Confidence interval



Question 15

Not answered

Marked out of 3.00

Statistical Tests 4

For each of the following vignettes choose the most appropriate statistical test/s.

Comparison of final outcome measures between two groups (placebo and antipsychotic) in a psychosis treatment study, taking into account significant baseline difference between groups. Results are normally distributed.

Comparison of final outcome measures between the three groups (CBT, SSRI, placebo) in a depression treatment study. Results are normally distributed.

Comparison of a measure of coping styles in a depressed and a control group. Results are continuous outcomes with skewed distributions.

Your answer is incorrect.

Explanation:

Q1: In question 1, the baseline differences can be considered as covariates while mean differences are sought between the two groups. ANCOVA will be appropriate for this purpose.

Q2: For mean comparisons across more than two groups, ANOVA is a handy method.

Q3: As the distribution is skewed, a non-parametric approach for comparing unpaired observations (i.e. Mann-Whitney U Test) can be used.

The correct answer is: Comparison of final outcome measures between two groups (placebo and antipsychotic) in a psychosis treatment study, taking into account significant baseline difference between groups. Results are normally distributed. → ANCOVA., Comparison of final outcome measures between the three groups (CBT, SSRI, placebo) in a depression treatment study. Results are normally distributed. → ANOVA., Comparison of a measure of coping styles in a depressed and a control group. Results are continuous outcomes with skewed distributions. → Mann-Whitney test.

Question 16

Not answered

Marked out of 3.00

Statistical Concepts 2

Find the appropriate measure from the list provided that corresponds to the description below.

The 'uncertainty' surrounding a mean or other summary estimate.

A measure reflecting the probability of making a false positive conclusion from the results.

The probability that your experiment will find a pre-specified difference when one truly exists.

Your answer is incorrect.

Explanation:

Q1: Standard error reflects how much sampling fluctuation a measurement can have by providing the uncertainty around the estimated value.

Q2: P value (generally 5%) is synonymous with type 1 error rate. It reflects the probability of making a false positive conclusion from the results.

Q3: In simplistic terms, statistical power is the odds that you will observe a treatment effect when it occurs (question 3).

The correct answer is: The 'uncertainty' surrounding a mean or other summary estimate. → Standard Error, A measure reflecting the probability of making a false positive conclusion from the results. → P-value, The probability that your experiment will find a pre-specified difference when one truly exists. → Power

Question 17

Not answered

Marked out of 3.00

Numbers in critical appraisal II

For each of the questions below, select the one most appropriate number from the list provided.

The point at which the confidence interval for an odds ratio becomes non-significant.

Odds for a 90% event rate.

Generally accepted usual risk, expressed as a percentage, of type I error in a randomised controlled trial.

Your answer is incorrect.

Explanation:

Q1. When the confidence interval range includes the value 0 for mean differences, the results become insignificant. The same applies to a range of ratios when they include the value 1.

Q2. The event rate is a probability. Odds = $[\text{probability}/(1-\text{probability})] = [90\%/(1-90\%)] = [0.9/(1-0.9)] = 0.9/0.1 = 9$

Q3. 5% error is the normally accepted type 1 error rate. This converts to a p-value of 0.05.

The correct answer is: The point at which the confidence interval for an odds ratio becomes non-significant. → 1, Odds for a 90% event rate. → 9, Generally accepted usual risk, expressed as a percentage, of type I error in a randomised controlled trial. → 5

Question 18

Not answered

Marked out of 5.00

Glossary of Statistical Terms 2

Identify the correct terminology used in statistics for each of the following definitions given below;

The middle value in a distribution with equal number of values on either side

The value that occurs most frequently in a distribution

The average of all the different values in a given set

A term that refers to 25% of the data in a distribution.

The difference between the 75th percentile and the 25th percentile

Your answer is incorrect.

Explanation: The **median** is the middle value with an equal number of values on either side. The median is the 50th percentile. The **mode** is the value that occurs most frequently. If there are 2 or more values that occur with equal frequency, the distribution is described as multi-modal. The **mean** is the average of all the different values. **Quartile** refers to 25% of the data. So, the first quartile includes values up to the 25th percentile. The **inter-quartile range** is the difference between the 75th percentile and the 25th percentile.

The correct answer is: The middle value in a distribution with equal number of values on either side → Median, The value that occurs most frequently in a distribution → Mode, The average of all the different values in a given set → Mean, A term that refers to 25% of the data in a distribution. → Quartile, The difference between the 75th percentile and the 25th percentile → Interquartile range

Question 19

Not answered

Marked out of 4.00

Glossary of Statistical Terms 3

Identify the correct terminology used in statistics for each of the following definitions given below:

The term used to describe a distribution whose values shows either a negative or a positive skew.

It is the square root of the variance

It is a measure of how close the sample mean is likely to be to the population mean.

The term used to describe a distribution in which the values are dispersed equally on either side of the mean.

Your answer is incorrect.

Explanation:

Parametric distribution is a distribution in which the values are distributed equally on either side of the mean. In a parametric distribution the mean = median = mode. **Non-parametric** distribution refers to any distribution other than a normal distribution. The distribution of values shows either a negative skew or a positive skew.

Standard deviation and variance are measured of the spread of values around the mean. The standard deviation (S.D.) is the square root of the variance. In a parametric / normal distribution, 68% of values lie within 1 S.D. on either side of the mean. About 95% of values lie within 2. S.D. (1.96 S.D., to be specific) on either side of the mean.

Standard error of the mean refers to the S.D. of a distribution of sample means. It is a measure of how close the sample mean is likely to be to the population mean.

The correct answer is: The term used to describe a distribution whose values shows either a negative or a positive skew. → Non-parametric, It is the square root of the variance → Standard deviation. It is a measure of how close the sample mean is likely to be to the population

it is the square root of the variance → standard deviation, it is a measure of how close the sample mean is likely to be to the population mean. → Standard error of the mean, The term used to describe a distribution in which the values are dispersed equally on either side of the mean. → Parametric



Question 20

Not answered

Marked out of 4.00

Plots & Graphs (1)

Using the definitions given below, identify the type of plots/graphs used in statistical illustrations:

A survival curve in which the survival probability is plotted against the time from baseline.

A graph showing the median, range and interquartile range of a set of values

A diagram showing the proportional distribution of a categorical variable using a divided circle.

A diagram that illustrates the distribution of a categorical variable by showing a separate horizontal or vertical bar for each 'category', its length being proportional to the number of observations in that 'category'

Your answer is incorrect.

Explanation:

1) Kaplan-Meier plot is a survival curve in which the survival probability is plotted against the time from baseline. It is used when exact times to reach the endpoint.

2) A box-and-whisker plot is a diagram illustrating the distribution of a variable; it indicates the median, upper and lower quartiles, and, often, the maximum and minimum values.

3) A pie chart is a diagram showing the frequency/proportional distribution of a categorical or discrete variable. A circular 'pie' is split into sections, one for each 'category'; the area of each section is proportional to the frequency in that category.

4) A bar or column chart is a diagram that illustrates the distribution of a categorical or discrete variable by showing a separate horizontal or vertical bar for each 'category', its length being proportional to the (relative) frequency in that 'category'.

(Ref: Medical statistics at a glance-Petrie & Sabin)

The correct answer is: A survival curve in which the survival probability is plotted against the time from baseline. → Kaplan-Meier plot, A graph showing the median, range and interquartile range of a set of values → Box and whisker plot, A diagram showing the proportional distribution of a categorical variable using a divided circle. → Pie Chart, A diagram that illustrates the distribution of a categorical variable by showing a separate horizontal or vertical bar for each 'category', its length being proportional to the number of observations in that 'category' → Bar chart



Question 21

Not answered

Marked out of 5.00

Plots & Graphs (2)

Using the definitions given below, identify the type of plots/graphs used in statistical

A diagram relating the pre-test probability of a diagnostic test result to the likelihood and the post-test probability.

A diagram used in a meta-analysis showing the estimated effect of each trial and the pooled average with confidence intervals

A diagram that illustrates the relative frequency distribution of a continuous variable by using connected bars.

The two-dimensional plot of one variable against another, with each pair of observations marked by a point

A mixture of a diagram and a table used to illustrate the distribution of data.

Your answer is incorrect.

Explanation:

- 1) Fagan's Nomogram (also called Bayesian Nomogram) is a diagram relating the pre-test probability of a diagnostic test result to the likelihood and the post-test probability. It is usually used to convert the former into the latter.
- 2). A forest plot is a diagram used in a meta-analysis showing the estimated effect in each trial and their average (with confidence intervals)
- 3). A Histogram is a graph of continuous data with the data categorized into many classes. It illustrates the (relative) frequency distribution of a continuous variable by using connected bars. The bar's area is proportional to the (relative) frequency in the range specified by the boundaries of the bar.
- 4) Scatter diagram is a two-dimensional plot of one variable against another, with each pair of observations marked by a point

5). A steam and leaf plot is a mixture of a diagram and a table used to illustrate the distribution of data. It is similar to a histogram and is effectively the data values displayed in increasing order of size.

(Ref: Petrie & Sabin, Medical statistics at a glance 1st edn.)

The correct answer is: A diagram relating the pre-test probability of a diagnostic test result to the likelihood and the post-test probability. → Fagan's Nomogram, A diagram used in a meta-analysis showing the estimated effect of each trial and the pooled average with confidence intervals → Forest plot, A diagram that illustrates the relative frequency distribution of a continuous variable by using connected bars. → Histogram, The two-dimensional plot of one variable against another, with each pair of observations marked by a point → Scatter plot, A mixture of a diagram and a table used to illustrate the distribution of data. → Steam and leaf plot



Question 22

Not answered

Marked out of 5.00

Sampling methods 5

Identify the type of sampling for each of the following descriptions;

Every individual in a population has an equal chance of being chosen.

Choose...

Dividing the population into clusters, typically on the basis of geography, and taking a random sample of the clusters

Choose...

Dividing the population into groups on the basis of some suspected confounding characteristic and then sampling each group to ensure fair distribution of these variables.

Choose...

Selecting the population on the basis of whoever is accessible

Choose...

Choosing every nth item from a list, beginning at a random point.

Choose...

Your answer is incorrect.

Explanation:

- 1) In random sampling every individual in a population has an equal chance of being chosen.
- 2) Cluster sampling involves dividing the population into naturally occurring clusters, typically on the basis of geography, and taking a random sample of the clusters (and thus including all of the subjects that reside within a cluster)
- 3) Stratified sampling involves dividing the population into groups on the basis of some suspected confounding characteristic and then sampling each group to ensure fair distribution of these variables.
- 4) Convenience sampling is selecting on the basis of convenience - not using a sampling frame but including whoever is easily accessible.
- 5) Systematic sampling involves choosing every nth item from a list sometimes beginning at a random point

or, systematic sampling involves choosing every n th item from a list, sometimes beginning at a random point.

The correct answer is: Every individual in a population has an equal chance of being chosen. → Random sampling, Dividing the population into clusters, typically on the basis of geography, and taking a random sample of the clusters → Cluster sampling, Dividing the population into groups on the basis of some suspected confounding characteristic and then sampling each group to ensure fair distribution of these variables. → Stratified sampling, Selecting the population on the basis of whoever is accessible → Convenience sampling, Choosing every n th item from a list, beginning at a random point. → Systematic sampling



Question 23

Not answered

Marked out of 4.00

Reliability

Identify the type of reliability for each of the following descriptions:

The level of agreement between two ratings on the same material by same rater but at different times

Choose...

The level of agreement between the ratings of the same material by two raters

Choose...

The level of agreement between one-half of a test with the other.

Choose...

The level of agreement between the Chinese version and the English version of a depression scale in the same group of bilingual subjects fluent in both languages.

Choose...

Your answer is incorrect.

Explanation:

- 1) Test-retest reliability describes the level of agreement between two ratings on the same material but at different times
- 2) Inter-rater reliability describes the level of agreement between the ratings on the same material by two raters. (2 or more assessors make simultaneous measurements)
- 3) Split half reliability describes the extent of agreement between one-half of a test with the other (e.g. a 20 item scale is divided into two halves of 10 items each)
- 4) Alternate form reliability describes the level of agreement between two different versions of the same test. The alternate-form reliability can also be used to devise modified versions of a test that helps to overcome the "practice effect" by changing the wording of the test questions in a functionally equivalent form.

The correct answer is: The level of agreement between two ratings on the same material by same rater but at different times → Test-retest reliability, The level of agreement between the ratings of the same material by two raters → Inter-rater reliability, The level of agreement between one-half of a test with the other. → Split-half reliability, The level of agreement between the Chinese version and the English version of a depression scale in the same group of bilingual subjects fluent in both languages. → Alternate form reliability



Question 24

Not answered

Marked out of 5.00

Statistical tests 6

Choose the most appropriate test for each of the following situations;

In a depression treatment study, the comparison of final outcome measures between the three groups (CBT, interpersonal therapy and placebo). The results are normally distributed

Investigation of the association between several risk factors and the risk of developing depression

The comparison of the risk of developing depression in 2 groups of 500 people. One group is comprised of people who regularly use alcohol and the other of people who do not regularly consume alcohol

Comparison of a measure of coping styles measured using a rating scale in a psychotic and a control group. The results are continuous outcomes with skewed distribution

Comparison of final outcome measure between 2 groups (placebo and antidepressants) in a depression treatment study taking into account significant baseline difference between groups. The results are normally distributed.

Your answer is incorrect.

Explanation:

1) One-way ANOVA (analysis of variance) is a parametric test used to compare 3 or more groups. The first analysis is to find out there is any difference across the groups.

2) Logistic regression is used in data sets where the dependent variable Y is binary in nature (e.g. diagnosis present/absent, dead/alive, etc.). The logistic regression model can be described as that which examines the relationship between one or more independent variables X1, X2, X3....Xk and the risk of developing an event Y.

3) A chi-square test is used in frequency data and is a test of association between 2 categorical variables. The categorical data from different samples are compared regarding any real differences. It is used for compare proportions between 2 or more groups

4) Mann-Whitney U test is the non-parametric equivalent of the 2 sample t test, comparing the distributions of two independent groups of observation. Mann Whitney U test is used to see whether there is a significant difference between 2 sets of data obtained from 2 different sets of subjects.

5) ANCOVA (analysis of covariance) is used to compare mean values of 2 groups that differ at baseline. It is valid only if data is normally distributed and if baseline and final measures are expected to be correlated.

The correct answer is: In a depression treatment study, the comparison of final outcome measures between the three groups (CBT, interpersonal therapy and placebo). The results are normally distributed → ANOVA, Investigation of the association between several risk factors and the risk of developing depression → Logistic regression, The comparison of the risk of developing depression in 2 groups of 500 people. One group is comprised of people who regularly use alcohol and the other of people who do not regularly consume alcohol → Chi-squared test, Comparison of a measure of coping styles measured using a rating scale in a psychotic and a control group. The results are continuous outcomes with skewed distribution → Mann -Whitney test, Comparison of final outcome measure between 2 groups (placebo and antidepressants) in a depression treatment study taking into account significant baseline difference between groups. The results are normally distributed. → ANCOVA



Question 25

Not answered

Marked out of 3.00

Multivariate techniques

Identify one test for each of the following descriptions

It reduces a large number of variables that are highly inter-correlated into a smaller group of representative variables

Choose...

It predicts the value of a single dependent variable based on the values of several independent variables

Choose...

It predicts the value of a dichotomous variable based on the values of several independent variables

Choose...

Your answer is incorrect.

Explanation:

1) **Factor analysis** is essentially a method of data reduction - here groups of variables that are highly inter-correlated in a dataset are identified. Then simple correlation analysis of individual factors is carried out. A good example is factor analysis of positive symptoms by Liddle to describe three groups of variables - reality distortion, psychomotor poverty and disorganization syndromes.

2) **Multiple linear regression** is a linear regression model in which there is a single dependent variable and 2 or more independent variables. In multiple linear regression, several independent variables together predict a single dependent variable. Consider $y = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + \dots + b_nx_n$. Multiple regression is a multivariate technique.

3) **Logistic regression** is used to predict the relationship between a binary dependent variable and several independent variables. For linear regression, the dependent variable must be continuous. If the dependent variable is binary, then the logistic regression is used. It is commonly used to get OR, RR or hazard ratio for independent variables that affect the dependent variable.

The correct answer is: It reduces a large number of variables that are highly inter-correlated into a smaller group of representative variables → Factor analysis, It predicts the value of a single dependent variable based on the values of several independent variables → Multiple regression, It predicts the value of a dichotomous variable based on the values of several independent variables → Logistic regression



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